ASWINI M S

7010917929 | aswini.ms2023eee@sece.ac.in | LinkedIn | Github | Portfolio | DOB: 27/05/2006

SUMMARY

Electrical and Electronics Engineering undergraduate specializing in AI-driven autonomous systems, computer vision, machine learning, and embedded intelligence. Experienced in developing scalable solutions for industrial automation, climate forecasting, and intelligent navigation using Python, Deep Learning, Edge AI, and modern software frameworks. Passionate about building innovative technologies that bridge artificial intelligence with real-world engineering applications.

EDUCATION

B.E (EEE) – Sri Eshwar College of Engineering | **CGPA : 9.26(upto 5th sem)** **2023-2027**

HSC – Jain Vidyalaya | **92.4%** **2021-2023**

SSLC – Jain Vidyalaya | **94.2%** **2020-2021**

INTERNSHIPS

Technology Intern – Global Technologies (July 2025)

Built and maintained data workflow automation pipelines using Apache Airflow, Apache Spark, and Apache NiFi for large-scale ETL processing. Gained hands-on exposure to end-to-end data flow, system reliability practices, and distributed processing across heterogeneous sources.

MERN Stack Intern – RV Tech (January 2025)

Developed full-stack web applications with MongoDB, Express.js, React, and Node.js; shipped RESTful APIs and responsive interfaces. Followed clean-architecture and scalability practices; collaborated on real-world deployment workflows.

PROJECTS

DUMPSIGHT — ***AI-driven drone-based smart dump packing system for autonomous mining truck*** Github | Link

Reduced autonomous dump spacing from 7.38 m to 4.05 m — a ~75% improvement in packing efficiency for mining trucks operating in confined dump yards. Engineered a YOLOv8-based perception pipeline with OpenCV for real-time obstacle detection, intelligent dump-point selection, and collision-free path planning. Designed a hybrid Edge AI + Cloud architecture for low-latency perception and decision-making, reducing simulated truck idle time across multi-truck dump cycles.

Technologies: *Python, YOLOv8, OpenCV, Computer Vision, Machine Learning, Autonomous Systems, Route Optimization, Path Replanning*

CLIMA-TWIN — ***AI-powered climate forecasting and risk platform*** Github | Link

Trained 38 per-district LSTM models delivering 7-day rainfall and temperature forecasts across Tamil Nadu, with Monte Carlo Dropout uncertainty bands for confidence-aware decisions. Built a multi-source fusion pipeline ingesting IMD, Open-Meteo, and MOSDAC INSAT feeds, with an async daemon refreshing twin state continuously. Implemented a downstream risk engine translating forecasts into flood, heatwave, and drought scores — methodology directly applicable to weather-aware fleet scheduling and equipment derating in extreme conditions

Technologies: *Python, PyTorch, LSTM, FastAPI, React, TypeScript, Leaflet, Time-Series Forecasting, Uncertainty Quantification, Data Fusion*

SAFEMAP — ***Crime analytics platform with natural-language SQL-Karnataka State Police Datathon 2026*** Github

Engineered an ingestion pipeline scraping opencity.in, reconciling district names against 2011 census data, and loading 5,000+ records into a DuckDB analytics layer. Integrated a Groq Llama 3.3 70B natural-language-to-SQL interface with schema-grounded prompts and SELECT-only safety validation, returning executed queries in under 500 ms. Implemented z-score anomaly detection and YoY trend analysis surfaced via an interactive deck.gl choropleth covering 31 districts and 1,100+ police units.

Technologies: *Python, FastAPI, DuckDB, React, deck.gl, Groq Llama 3.3 70B, Natural Language to SQL, Geospatial Analytics, Anomaly Detection*

VISIONMATE — ***Edge AI navigation system with LiDAR + camera sensor fusion***

Developing an Edge AI navigation platform on Jetson Orin Nano with YOLOv8 and RPLidar for real-time obstacle detection and voice-guided assistance. Integrating LiDAR mapping, intelligent path planning, and ESP32-based sensor control for cloud-independent navigation. Leveraging an industrial-grade edge AI stack directly applicable to autonomous heavy-equipment perception and navigation.

Technologies: *Python, PyTorch, YOLOv8, OpenCV, Jetson Orin Nano, RPLidar, ESP32, Arduino, Computer Vision, Edge AI, LiDAR*

ACHIEVEMENTS

Caterpillar Tech Challenge Finalist: Built an AI-based autonomous mining truck system featuring obstacle detection, emergency response, and dynamic path replanning.

NASA Space Apps Challenge: Selected among the top 947 teams globally for developing Clima360, a climate awareness platform visualizing 20 years of global data, recognized for impactful design and data-driven storytelling.

Freshathon: Shortlisted among the Top 10 teams out of 250 and secured 1st place at the department level, awarded for problem-solving, innovation, and teamwork.

Zeroth-Day Hackathon: Secured 3rd place at the college level, shortlisted among the top 50 teams out of 85 for building an impactful technical solution.

Competitive Programming: Achieved a highest LeetCode global contest rating of 1735 with 175+ problems solved; secured 4 STAR in C, 3 STAR in Java, and 1 STAR in SQL on HackerRank with multiple badges and certifications | Hackerrank | Leetcode

CERTIFICATIONS

Introduction to Internet of Things (IoT) | NPTEL

Mastering Data Structures and Algorithms using C and C++ | Udemy

Learn JAVA Programming | Udemy

Smart Grid: Basics to Advanced Technology | NPTEL

SQL(Basics and Intermediate) | Hackerrank

Problem Solving Basics | Hackerrank

Data Visualization with Power BI | Great Learning

SKILLS

Programming Languages: Python, C, Embedded C (ESP32 / Arduino), Java

Artificial Intelligence: PyTorch, Scikit-learn, OpenCV, YOLOv8, Computer Vision, Deep Learning, LSTMs, Time-Series Forecasting, MC Dropout

Data Engineering: Apache Airflow, Apache Spark, Apache NiFi, ETL Pipelines

Data & Visualization: Power BI, NumPy, Pandas, Plotly, deck.gl, Data Visualization

Edge Computing & Embedded AI: Jetson Orin Nano, ESP32, Arduino, RPLidar, LiDAR + Camera Sensor Fusion, Real-Time Systems

Databases & Technical Tools: SQL, DuckDB, MongoDB, Firebase, FastAPI, Git, GitHub, Linux (Basics), VS Code

Soft Skills: Communication, Problem Solving, Critical Thinking, Leadership, Teamwork, Time Management